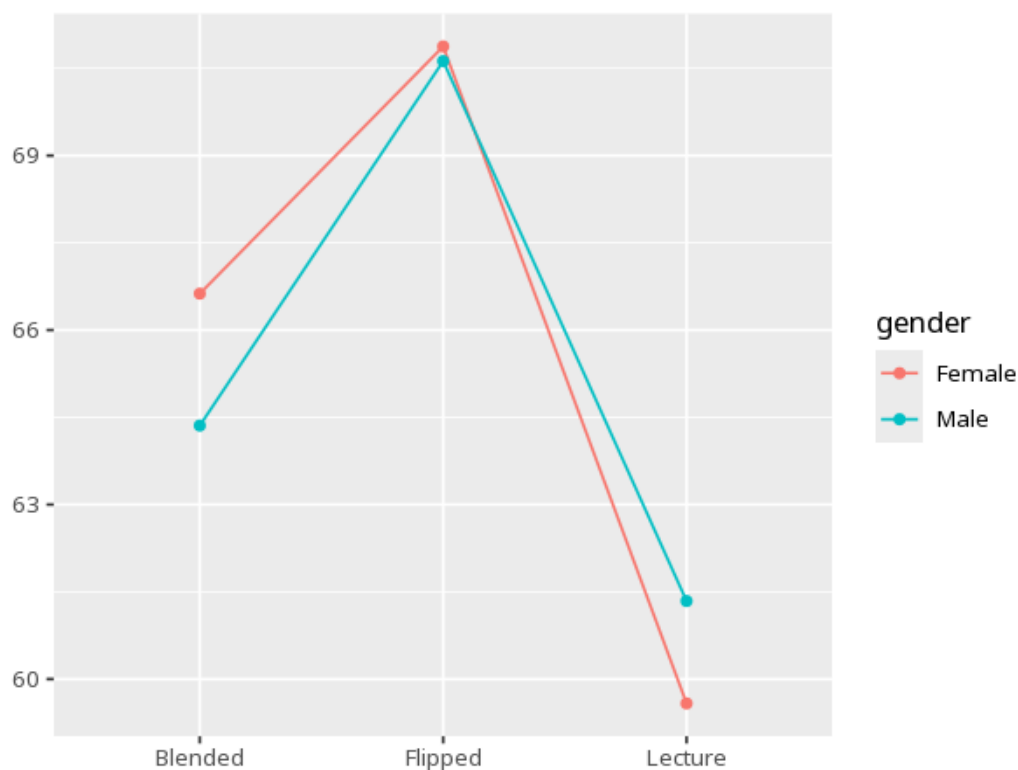


# 1 Factorial ANOVA

| Factor A $\times$ B     | N  | Mean  | Std. Dev. |
|-------------------------|----|-------|-----------|
| Blended $\times$ Female | 40 | 66.63 | 6.73      |
| Flipped $\times$ Female | 40 | 70.88 | 6.55      |
| Lecture $\times$ Female | 40 | 59.58 | 7.45      |
| Blended $\times$ Male   | 40 | 64.36 | 6.86      |
| Flipped $\times$ Male   | 40 | 70.62 | 6.69      |
| Lecture $\times$ Male   | 40 | 61.34 | 7.67      |

| Source                          | SS      | df | MS      | F     | p     | partial $\eta^2$ [95% CI] |
|---------------------------------|---------|----|---------|-------|-------|---------------------------|
| teaching_method                 | 4234.40 | 2  | 2117.20 | 43.14 | <.001 | 0.27 [0.19, 1.00]         |
| gender                          | 3.85    | 1  | 3.85    | 0.08  | .780  | 0.00 [0.00, 1.00]         |
| teaching_method $\times$ gender | 162.25  | 2  | 81.12   | 1.65  | .194  | 0.01 [0.00, 1.00]         |

| Contrast          | Mdiff | SE   | padj  | 95% CI         |
|-------------------|-------|------|-------|----------------|
| Blended - Flipped | -5.26 | 1.11 | <.001 | [-7.87, -2.64] |
| Blended - Lecture | 5.03  | 1.11 | <.001 | [2.42, 7.64]   |
| Flipped - Lecture | 10.29 | 1.11 | <.001 | [7.68, 12.90]  |



Each factor's F/p is its main effect; the A $\times$ B row tests whether the effect of

one factor depends on the other. A significant interaction usually takes priority - read it from the interaction plot before the main effects. A significant effect for any factor with 3+ levels tells you the levels differ somewhere, not which ones - use the post-hoc / simple-effects (emmeans) table to find the specific pairs. Your significance threshold ( $\alpha$ ) is 0.05.

A two-way ANOVA gave A×B interaction  $F(2,234)=1.65$ ,  $p = .194$ , partial  $\eta^2=.01$  [.00, 1.00].

Statistical basis: Factorial ANOVA (main effects + interaction) (Fisher, R. A. (1925). *Statistical Methods for Research Workers*. Oliver & Boyd.); Estimated marginal means for interaction / simple-effects follow-up (Lenth RV (2024). *emmeans*: Estimated Marginal Means, aka Least-Squares Means. R package.);  $\eta^2$  effect-size benchmarks (Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Routledge.).